Trust-Aware Continual Learning for IoT Intrusion Detection

Authorizing drift-triggered model updates with traffic evidence, an out-of-band reputation signal, and quarantined, reversible updates

Project report (report_ids)
VIT Project-I, Review 2 proof-of-concept

Repository: github.com/akarshdixit/trust-ids-poc

Contents: 1 Problem and goal - 2 System structure - 3 Repository layout - 4 Datasets - 5 Algorithms - 6 Experimental setup - 7 Results - 8 Robustness and the quarantine extension - 9 Related work and novelty - 10 Software, packages and backend - 11 Visualizations - 12 Testing - 13 How to run - 14 Limitations and future work - References

1. Problem and goal

An IoT intrusion detection system (IDS) deployed on a changing network has to keep learning: devices are added, firmware changes and benign traffic drifts. Continual learning makes that possible, but it also opens an attack surface. A patient attacker can shape traffic so that it drifts slowly from attack towards benign; if the IDS adapts to every detected drift, it gradually learns that the attack is normal (*slow poisoning*, the 'boiling frog' attack).

The goal of this project is an **authorization gate** that sits between drift detection and the model update. Every proposed update must be authorized. The gate combines two kinds of evidence:

- **Traffic evidence** computed from the traffic window itself (attribution consistency, topology diversity, temporal stability). It is cheap but attacker-influenceable.
- **An out-of-band reputation signal** (where the traffic goes: destination reputation or C2 cost), which is independent of the traffic features and therefore not controlled by the attacker.

This version adds a **quarantine extension**: the reputation channel is treated as sequential, fallible evidence about a whole drift episode; updates are applied provisionally and rolled back when the evidence turns against them. The research question is whether such a gate keeps the benefits of continual learning on benign drift while refusing poisoning, including when the reputation feed is noisy or late.

Honest scope. No public IoT dataset carries a reputation field, so the signal is simulated from ground-truth origin (benign 1.0, attack 0.05) and then degraded on purpose. The claim is what an authorization mechanism achieves *once it has an independent channel*, not that the gate detects attackers by itself.

2. System structure

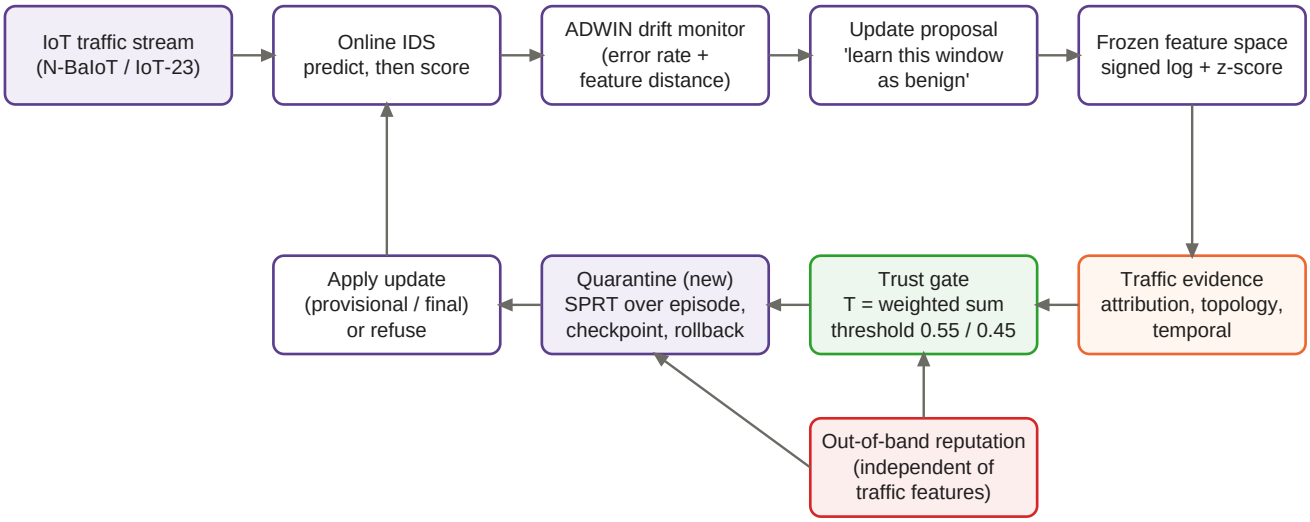


Figure 1. Pipeline. Every arm sees the same stream and the same drift trigger; the arms differ only in how they decide whether to apply a proposed update.

Per 20-record round:

- **Prequential evaluation.** Every arm predicts each record first and is scored afterwards (test-then-train), so no arm ever sees a label before predicting it.
- **Drift trigger.** A shared ADWIN detector watches the frozen static arm's error rate (optionally also each record's distance from the calibration centre). A change point opens a 15-round drift-active

window.

- **Proposal.** Only in drift-active rounds is an update proposed: 'learn this window as benign'. Outside it nobody learns.
- **Decision.** Each arm's policy (never, always, gate, EWC, quarantine) decides; authorized updates are applied with River's learn_one.

Six arms run side by side, which makes every comparison a controlled experiment:

Arm	Policy
Static (never updates)	Trained on calibration data once, never updates. Reference for 'no adaptation'.
Ungated (learns everything)	Applies every drift-triggered proposal.
Traffic-only gate	Traffic evidence only, no reputation veto (stands in for ExpIDS-style drift gating).
Full gate (one-shot veto)	Traffic evidence plus a one-shot reputation veto (the Review-2 mechanism).
EWC (lambda = 50)	Applies every proposal, with Elastic Weight Consolidation anchoring weights to calibration. Tests whether the gate's benefit is merely 'not forgetting'.
Quarantine gate (new)	Sequential reputation test over the drift episode, provisional updates, rollback (section 5.6).

3. Repository layout

Path	Role
models/online_ids.py	OnlineIDS (StandardScaler | LogisticRegression, ADWIN) and EWCOnlineIDS
evidence/featurespace.py	Frozen signed-log + z-score space (trust boundary)
evidence/extract.py	Attribution, topology and temporal evidence
gate/trust_gate.py	One-shot gate: weighted trust score + reputation veto
gate/quarantine.py	SPRT, provisional updates, checkpoints, rollback (new)
data/load_nbaiot.py	N-BaIoT loader and the two scenario streams
data/load_iot23.py	IoT-23 Zeek conn-log parser and streams (new)
data/synth_stream.py	Synthetic 20-feature streams (regression check)
data/reputation.py	Imperfect reputation feed: noise, flips, lag (new)
data/fake_nbaiot.py	Fake N-BaIoT-format files for tests (new)
experiments/run_scenarios.py	Orchestrator; writes results/*_log_<tag>.json
experiments/streams.py	build_stream(): one stream builder for CLI, API, app
experiments/sweeps.py	Seeds, reputation, EWC and quarantine sweeps (new)
experiments/make_plots.py	Static figures per data source
backend/server.py, console.html	FastAPI backend, SSE logs, live console (new)
app.py, charts.py, equations.py	Streamlit demo, Altair charts, equations
tests/	108 pytest tests
results/	Logs (.json) and figures (.png)
PROJECT_GUIDE.txt	Changes, how to run, meaning of every chart

4. Datasets

4.1 N-BaIoT (primary)

UCI dataset 442 (Meidan et al., 2018): traffic of nine commercial IoT devices infected with Mirai and BASHLITE (Gafgyt). Each row is a packet window summarised into 115 statistical features (packet size, jitter, counts, covariance and correlation over 5 time windows, for host, channel and socket aggregations), so no packet parsing is needed. Only the files the pipeline uses were downloaded:

File	Rows x cols	Used for
Danmini_Doorbell/benign_traffic.csv	49,548 x 115	benign traffic, both scenarios
Danmini_Doorbell/mirai_attacks/scan.csv	107,685 x 115	attack traffic, Scenario 2
Danmini_Doorbell/gafgyt_attacks/combo.csv	59,718 x 115	sanity check (check_before_real_data.py)
Ennio_Doorbell/benign_traffic.csv	39,100 x 115	second device, Scenario 1

All files: 0 NaN, 0 Inf, identical column order. 20 columns (covariance and Pearson-correlation features) take negative values, and the dynamic range spans about 19 orders of magnitude (maximum 5.7e17), which is why evidence is computed in a calibrated space (section 5.2).

4.2 IoT-23 (second, independent dataset)

Stratosphere Laboratory, CTU Prague (Garcia et al., 2020). Only the labelled Zeek flow logs (conn.log.labeled) were downloaded, not the PCAPs. The parser handles the format quirks (space-separated last columns, 'det_label' vs 'detailed-label' headers, label case, '-' for missing values).

Capture	Flows	Role
CTU-IoT-Malware-Capture-34-1	23,145 (21,222 Mirai: DDoS + C&C;)	attack, plus device-A benign (1,923 flows)
CTU-Honeypot-Capture-5-1	1,374 benign (Amazon Echo)	device B, Scenario 1
CTU-Honeypot-Capture-4-1	452 benign (Philips HUE)	parsed, too small to use

21 features per flow: 8 numeric (duration, originator/responder bytes and packets, IP bytes, missed bytes), 3 protocol one-hots (tcp, udp, icmp) and 10 connection-state one-hots (S0, SF, REJ, ...).

4.3 Synthetic stream and other sources

- **Synthetic:** 20 Gaussian features (data/synth_stream.py), used as a fast regression check with known ground truth.
- **Fake N-BaIoT:** generated files with the real column names and dynamic range, only for tests on machines without the data.
- **CICIoT2023** was considered but not used: UNB requires a registration form.

4.4 The simulated reputation channel

meta['reputation'] is 1.0 for benign origin and 0.05 for attack origin. data/reputation.py degrades it in three independently controllable ways: Gaussian noise (sigma), a per-record flip rate (the feed reports the wrong verdict, as with a stale or poisoned blocklist) and lag (the verdict for record i arrives at record i + lag). Features, labels and phases are untouched, so every comparison against the oracle run stays controlled.

5. Algorithms

5.1 Online IDS and drift detection

The detector is River's pipeline StandardScaler | LogisticRegression, trained one record at a time with SGD. ADWIN (adaptive windowing, Bifet and Gavalda 2007) keeps a variable-length window of a 0/1 error stream and signals a change when two sub-windows have significantly different means. The shared monitor uses the frozen static arm, whose error rate is an unbiased measure of how far live traffic has moved from calibration. The optional second trigger runs ADWIN on each record's normalised distance from the calibration centre, ||z - c|| / sqrt(d), which catches benign-to-benign drift that never raises the error rate.

5.2 Calibrated feature space (trust boundary)

```
z = (slog(x) - mu_calib) / sd_calib        slog(x) = sign(x) * log(1 + |x|)
mu_calib, sd_calib: fit ONCE on calibration benign records, then frozen
```

The evidence signals are geometric, so features must be on comparable scales. On raw N-BaIoT values temporal stability evaluates to 1.7e-17 and five jitter variance columns carry 100% of every cosine, so nothing is ever authorized. The signed log keeps the sign of correlation features. Fitting on calibration data only means the attacker can never move the reference frame.

5.3 Traffic evidence

```
shift        = mean(window_z) - trusted_baseline
attribution  = (cos(shift, reference_direction) + 1) / 2      (0.5 if none yet)
reference_direction <- 0.7 * reference_direction + 0.3 * shift  (after authorization)
topology     = H(histogram of 5 features, 10 bins) / log(9)     (normalised entropy)
temporal     = 1 / (1 + ||mean(window_z) - mean(previous_window_z)||)
trust score  T = 0.4 * attribution + 0.3 * topology + 0.3 * temporal
```

Attribution asks whether this drift continues a direction that was already authorized; topology rewards varied, device-like behaviour over a narrow repeated channel; temporal stability rewards gradual change, which is exactly what a slow attacker maximises, so it is not sufficient on its own. Weights are normalised to sum to 1 (negative or all-zero weights are rejected).

5.4 One-shot trust gate (Review 2)

```
r = mean reputation of the 20-record window
if r < 0.3:            REFUSE  (out-of-band veto)
elif T >= 0.55:        AUTHORIZE
else:                  REFUSE  (weak traffic evidence)
```

5.5 Elastic Weight Consolidation baseline

After calibration the logistic-regression weights w* are frozen and the diagonal Fisher information F is estimated on calibration inputs (p(1-p) z^2, normalised to a maximum of 1). After each SGD step every weight is pulled back: w <- w - min(1, eta * lambda * F) (w - w*). This penalises moving important weights, the standard defence against catastrophic forgetting. Only the LR weights are anchored; the scaler keeps adapting.

5.6 Quarantine extension (new)

Sequential reputation test. Each record's reputation is a vote (benign if > 0.5). The feed is assumed wrong on at most e = 30% of records, so under 'benign episode' a benign vote has probability 1 - e and under 'attack episode' probability e. Wald's sequential probability ratio test accumulates the log-likelihood ratio over the whole drift episode:

```
step  = ln((1 - e) / e) = 0.847                per vote: LLR += step (benign) or -= step
A     = ln((1 - beta) / alpha) = +4.60         upper bound; LLR is clamped at A
B     = ln(beta / (1 - alpha)) = -4.60         lower bound; reaching B condemns the episode
alpha = beta = 0.01                            target false-condemn / miss rates
```

Clamping at A stops a long benign prefix (a lagged feed) from banking credit an attack would first have to burn off. Condemnation is sticky until a new episode starts.

Provisional updates, hold and rollback.

```
for each round:
    commit updates older than the hold (20 rounds)
    feed this window's votes to the SPRT (if drift-active or updates are held)
    if episode condemned and updates are held:
        restore the checkpoint taken before the episode's earliest held update
        replay held updates of other episodes on top (fresh checkpoints)
    if not drift-active:       nothing is proposed
    elif condemned:            refuse
    elif LLR <= 0:             hold (evidence not yet favourable)
    elif T >= 0.45:            save checkpoint, apply update PROVISIONALLY
    else:                      refuse
a new drift run starts a new episode only when nothing is held
```

Because a wrong acceptance is now reversible, the provisional threshold is 0.45, 0.1 below the one-shot gate's 0.55 (set before seeing results). The hold was first 5 rounds; the 600-record-lag sweep showed that a hold shorter than the feed's lag lets poison commit before the evidence arrives, so it was raised to 20 rounds (400 records). That is the only setting changed after seeing results, and the 5-round hold is kept as an ablation.

6. Experimental setup

Every stream begins with 400 labelled calibration records (30% attack) on which all arms are trained identically. The rest is processed in 20-record rounds.

Scenario	Phases (records)	What a good system does
2 - Slow poisoning	pre-attack benign (400); slow poison: real attack rows interpolated towards real benign rows in signed-log space (600); post-poison: attack held in the benign region (400); recovery check: raw attack again (200)	Refuse the poisoning updates, so the obvious repeat attack is still detected (recovery-check accuracy 1.0).
1 - Genuine drift	device A benign (400); gradual blend A to B (600); device B benign (400)	Accept the benign drift, so accuracy after the shift stays high.

Metrics. Prequential accuracy and F1 (attack = positive), and the windowed accuracy in each scenario's decisive phase: *recovery_check* for Scenario 2 and *post_drift* for Scenario 1. The sweeps also count *poisoned updates kept*: authorized updates whose window contained attack records and that were never rolled back. F1 is 0 in Scenario 1 because that scenario has no attack records after calibration; accuracy is the metric there.

Seeds. Seeds reshuffle calibration rows (N-BaIoT, IoT-23) or regenerate the world (synthetic). The default run uses the loaders' fixed seed 42.

7. Results

7.1 Slow poisoning: accuracy on the obvious repeat attack

Arm	N-BaIoT	IoT-23	Synthetic
Static (never updates)	1.000	1.000	1.000
Ungated (learns everything)	0.200	0.000	0.200
Traffic-only gate	0.200	0.300	0.200
Full gate (one-shot veto)	1.000	1.000	1.000
EWC (lambda = 50)	1.000	1.000	1.000
Quarantine gate (new)	1.000	1.000	1.000
Drift-active rounds	24	32	37

Table 1. Recovery-check accuracy, oracle reputation feed, seed 42.

Learning from every drift (ungated) or trusting traffic evidence alone (traffic-only) lets the attacker teach the model that the attack is normal: on real N-BaIoT both detect only 20% of the repeat attack, on IoT-23 0% and 30%. Both reputation-aware gates refuse every poisoning update and keep 1.000. EWC also survives here, but at a cost shown in 7.2.

Arm	N-BaIoT	IoT-23	Synthetic
Static (never updates)	1.000 +/- 0.000	1.000 +/- 0.000	1.000 +/- 0.000
Ungated (learns everything)	0.200 +/- 0.000	0.000 +/- 0.000	0.170 +/- 0.035
Traffic-only gate	0.200 +/- 0.000	0.265 +/- 0.032	0.171 +/- 0.035
Full gate (one-shot veto)	1.000 +/- 0.000	1.000 +/- 0.000	1.000 +/- 0.000
EWC (lambda = 50)	0.897 +/- 0.155	1.000 +/- 0.000	1.000 +/- 0.000
Quarantine gate (new)	1.000 +/- 0.000	1.000 +/- 0.000	1.000 +/- 0.000

Table 2. Recovery-check accuracy over 10 seeds, mean +/- 95% CI.

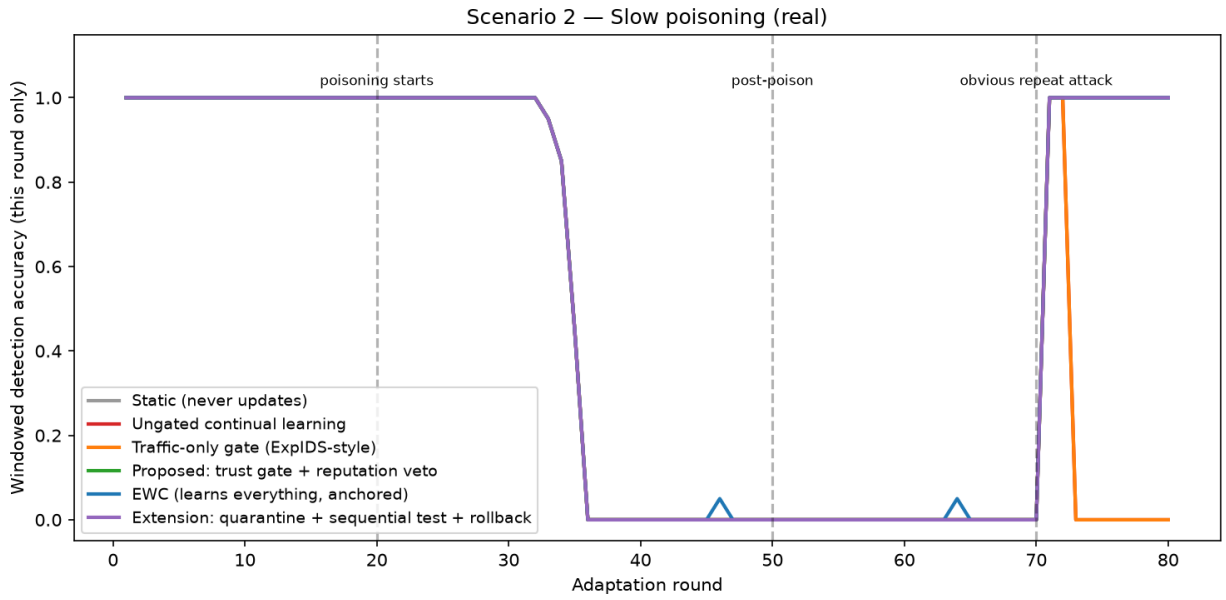


Figure 2. Scenario 2 on real N-BaIoT: windowed accuracy per round. Only the final 'obvious repeat attack' phase separates the arms.

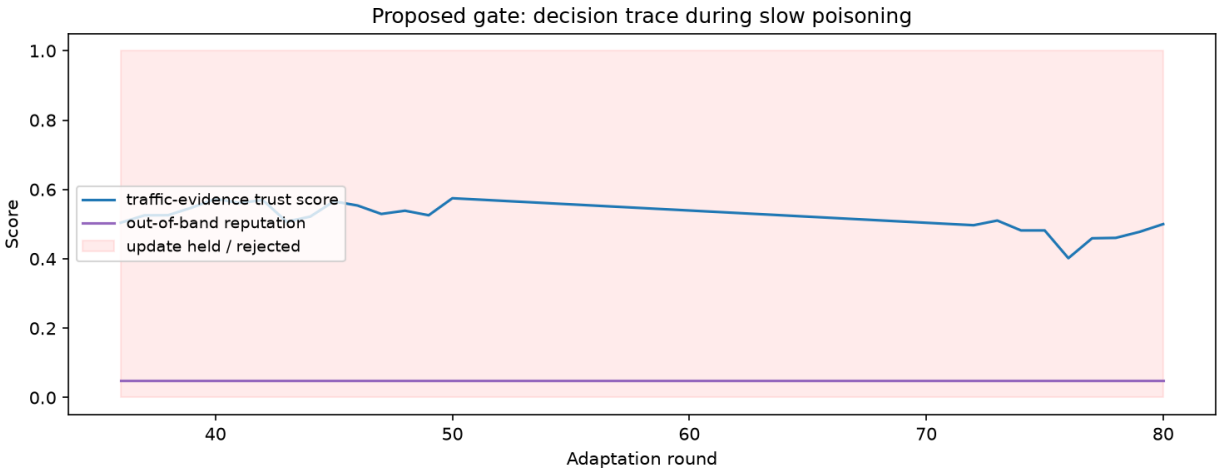


Figure 3. One-shot gate decision trace (drift-active rounds): the traffic trust score (blue) looks acceptable while reputation (purple) stays low, so the veto refuses (red shading).

7.2 Genuine drift: accuracy after a benign device shift

Arm	N-BaIoT + FD	IoT-23	IoT-23 + FD	Synthetic
Static (never updates)	0.993	0.860	0.860	0.000
Ungated (learns everything)	0.995	0.978	0.988	1.000
Traffic-only gate	0.993	0.978	0.905	1.000
Full gate (one-shot veto)	0.993	0.978	0.905	1.000
EWC (lambda = 50)	0.915	0.760	0.760	0.003
Quarantine gate (new)	0.995	0.975	0.975	1.000
Drift-active rounds	15	15	34	30

Table 3. Post-drift accuracy (FD = feature-drift trigger on). Without FD, real N-BaIoT detects no drift at all: a null result, all arms 0.993.

On IoT-23 the frozen static model falls to 0.860 after the device shift, while the gated learners recover to about 0.98: adaptation is worth having. EWC pays for its poisoning resistance by not adapting (0.760). With the feature-drift trigger, the one-shot gate refuses most real benign drift (IoT-23 0.905; on N-BaIoT its trust score peaks at 0.543, just under 0.55), while the quarantine gate's lower provisional threshold accepts it (0.975 on IoT-23, 0.995 on N-BaIoT).

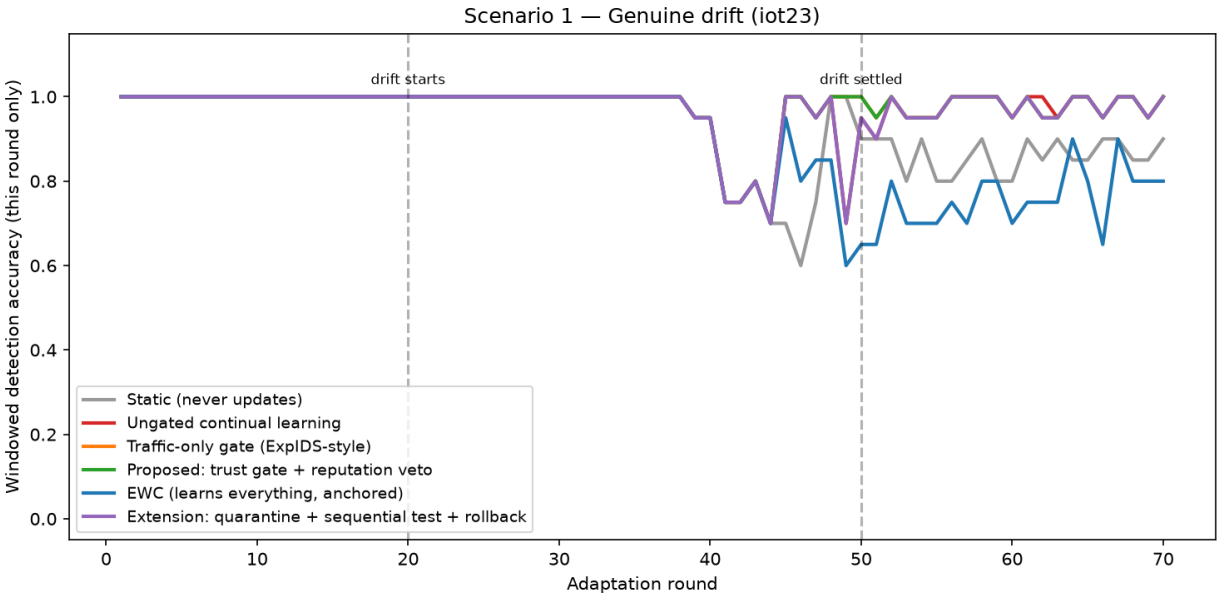


Figure 4. Scenario 1 on IoT-23: static drops after the shift; learners that adapt keep high accuracy, EWC does not.

8. Robustness and the quarantine extension

8.1 One-shot veto under an imperfect feed

Full gate under an imperfect reputation channel

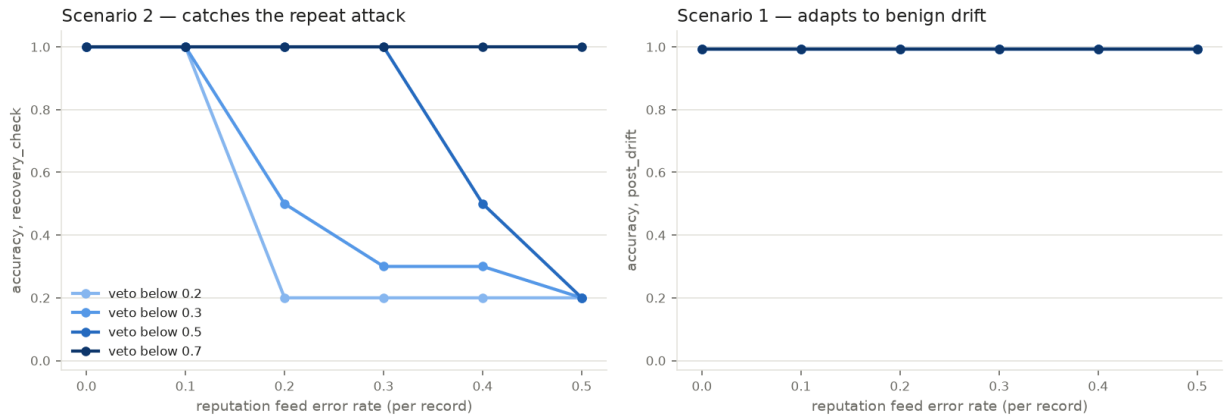


Figure 5. One-shot full gate vs feed error rate, one line per veto threshold (real N-BaIoT).

The default veto (0.3) holds only up to 10% feed error: at 20% the attack window's mean reputation crosses 0.3 often enough for poison to get in (0.50 on N-BaIoT, 0.60 on IoT-23). A 0.5 veto holds to 30-40%; 0.7 holds longer but starts blocking benign drift. A fixed per-window threshold is fragile.

8.2 EWC strength

EWC trades adaptation for stability; the gate needs neither trade

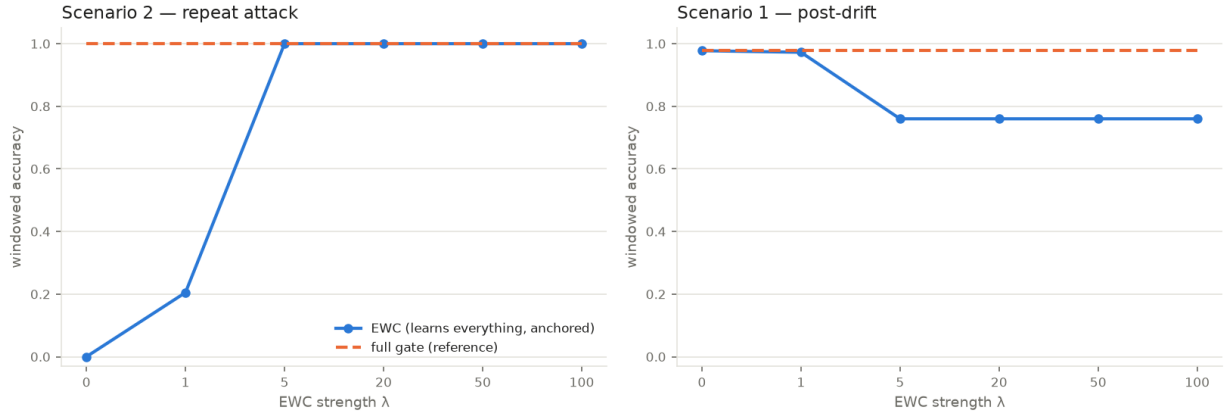


Figure 6. EWC lambda sweep on IoT-23: survives poisoning from lambda = 5 but post-drift accuracy falls to 0.76, vs 0.98 for the gate.

8.3 Quarantine vs one-shot gate

Feed	Metric	veto<0.3	veto<0.5	quarantine	no rollback	5-round hold
20% wrong	repeat-attack acc.	0.50 / 0.60 / 0.50	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00
	poison kept	2 / 6 / 3	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0
30% wrong	repeat-attack acc.	0.30 / 0.50 / 0.30	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00
	poison kept	3 / 5 / 7	1 / 1 / 1	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0
40% wrong	repeat-attack acc.	0.30 / 0.50 / 0.30	0.50 / 1.00 / 0.50	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00
	poison kept	3 / 6 / 9	2 / 2 / 2	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0

Feed	Metric	veto<0.3	veto<0.5	quarantine	no rollback	5-round hold
300 late	poison kept	0 / 1 / 0	0 / 1 / 0	0 / 0 / 0	0 / 4 / 0	0 / 0 / 0
600 late	poison kept	1 / 1 / 2	1 / 1 / 2	0 / 0 / 0	1 / 14 / 6	1 / 14 / 5

Table 4. Seed 42; each cell is N-BaIoT / IoT-23 / Synthetic. 'Poison kept' = authorized updates on attack windows never rolled back. Read from results/sweep_quarantine_*.json.

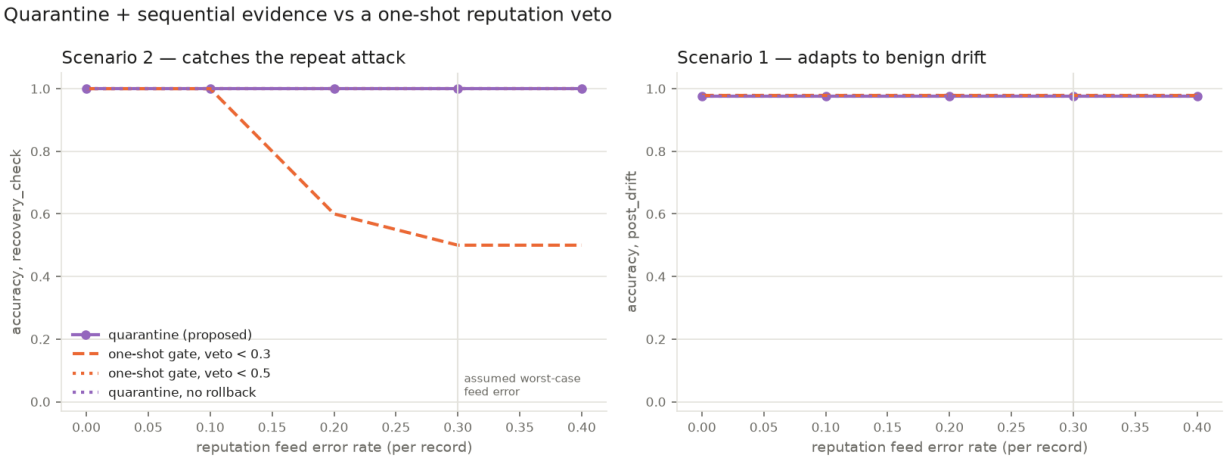


Figure 7. Quarantine sweep on IoT-23 (single seed).

Over 5 seeds (measured separately, benign drift with the feature-drift trigger on the real datasets):

Feed error	Repeat attack: quarantine vs one-shot (N-BaIoT / IoT-23 / synth)	Benign drift: quarantine vs one-shot
20%	1.000 / 1.000 / 1.000 vs 0.42 / 0.62 / 0.40	0.995 / 0.976 / 1.000 vs 0.997 / 0.960 / 1.000
30%	1.000 / 1.000 / 1.000 vs 0.22 / 0.32 / 0.20	0.995 / 0.976 / 1.000 vs 0.997 / 0.960 / 1.000
40%	0.88 / 1.000 / 1.000 vs 0.22 / 0.32 / 0.20	0.996 / 0.899 / 0.398 vs 0.997 / 0.960 / 1.000

Table 5. 5-seed means, hand-entered from a separate run.

Reading the ablations. The sequential test is what makes the gate robust to a *wrong* feed (zero poison at 20-40% error, where the one-shot veto at 0.3 keeps 2-9 updates and even 0.5 keeps 1-2). Rollback with a hold longer than the feed lag is what handles a *late* feed (without rollback 4-14 poisoned updates stay on IoT-23). The lower provisional threshold is what admits real benign drift. **Limit:** above the assumed 30% feed error the SPRT's guarantee is gone; it starts condemning benign drift (synthetic 0.40) and on N-BaIoT one seed in five lets the attack through.

9. Related work and novelty

A targeted (not systematic) literature search found every building block already published:

Idea	Prior work
Slow ('boiling frog') poisoning of self-updating detectors	Rubinstein et al., ANTIDOTE (IMC 2009); 'Can't Boil This Frog' (2020)
Poisoning that fakes or hides concept drift	Korycki and Krawczyk, adversarial concept drift detection (2020)
Drift detection and adaptation for security classifiers	Transcend (USENIX Sec 2017), CADE (USENIX Sec 2021), OWAD (NDSS 2023), INSOMNIA (2021), ExpIDS (2025)
Reject updates a trusted reference contradicts	RONI (Nelson et al. 2008); FLTrust trusted root data (NDSS 2021)
Trust-weighted updates and rollback after poisoning	Trust-aware federated IDS (AGAT-FL, 2025); Crab (2024), FedRecover (2023)
Sequential tests over IDS feedback	Zhu et al., collaborative IDS with sequential hypothesis testing (2010)

What was not found: a single, non-federated, self-labelling online IoT IDS that (a) gates its own drift-triggered updates on an out-of-band channel independent of the traffic features, and (b) treats that channel as sequential, fallible evidence about a whole drift episode, with explicit assumption-stated error rates, applying updates provisionally and rolling them back when the evidence turns. Federated work weighs *clients*; RONI and FLTrust need trusted *labelled* data; drift-adaptation work (OWAD, CADE, Transcend) does not model an adversary who manufactures the drift.

The defensible novelty claim is therefore the **combination** and the **episode-level sequential treatment of the independent channel**, not any single component.

10. Software, packages and backend

10.1 Packages (requirements.txt, pinned)

Package	Version	Used for
Python	3.12	runtime (virtual environment .venv)
river	0.26.1	online logistic regression, StandardScaler, ADWIN, metrics
numpy	2.5.4	vectors, evidence geometry
pandas	3.0.6	CSV / Zeek log loading, tables in the app
scipy	1.18.1	t-distribution confidence intervals in sweeps
matplotlib	3.11.2	static figures in results/
streamlit	1.66.0	interactive demo app
altair	6.3.0	interactive charts in the app
fastapi	0.144.0	backend REST API and SSE streams
uvicorn[standard]	0.54.0	ASGI server for the backend
httpx	0.28.1	app-to-backend client, API tests
pytest	9.1.1	test suite
reportlab	4.2.5	building this PDF (docs/build_report.py)

10.2 Backend

backend/server.py is a FastAPI application. Runs execute in worker threads; every round is logged to the console, logs/backend.log and an in-memory ring buffer that /api/logs/stream tails. Finished runs are saved to results/runs/ and reloaded on restart. Requests are validated with Pydantic (weights not all zero, feed error in (0, 0.5), ...). TRUST_IDS_TOKEN enables API-key auth (X-API-Key header or ?token=); TRUST_IDS_MAX_RUNS (default 2) caps concurrent runs and returns HTTP 429 above it. Port 8765 is used because 8000 is taken on the development machine.

Endpoint	Purpose
GET /	live log console
GET /api/health	status, dataset availability, auth, run limits
GET /api/results/{scenario}/{tag}	precomputed log (real, iot23, synth)
POST /api/runs	start a run: scenario, source, gate, reputation_noise, quarantine, seed, feature_drift, EWC, pace
GET /api/runs, /api/runs/{id}	list runs / run status and events
GET /api/runs/{id}/events	Server-Sent Events, one per round
GET /api/logs/stream	Server-Sent Events of all server logs

11. Visualizations

Colours are consistent everywhere: grey static, red ungated, orange traffic-only, green full gate, blue EWC, purple quarantine. PROJECT_GUIDE.txt explains each in more detail.

Visualization	What it shows / how to read it
scenario2_slow_poisoning_<tag>.png	Windowed accuracy per round in Scenario 2. Judge the last phase (obvious repeat attack): poisoned systems score low, protected ones 1.0.
scenario1_genuine_drift_<tag>.png	Same for the benign device shift; adapting systems stay high, static may drop.
gate_decision_trace_<tag>.png	Trust score (blue) and reputation (purple) on drift-active rounds; red shading = update held or refused.

Visualization	What it shows / how to read it
sweep_reputation_<tag>.png	One-shot gate vs feed error, one line per veto threshold (light to dark = low to high). Left: attack resistance, right: benign adaptation.
sweep_ewc_<tag>.png	EWC lambda vs both scenarios, gate as dashed reference: EWC trades adaptation for stability.
sweep_quarantine_<tag>.png	Quarantine (purple) vs one-shot gates (orange dashed 0.3, dotted 0.5) and the no-rollback ablation, vs feed error; grey line = assumed 30%.
App: per-arm columns	Records correct in this window and running accuracy, with change since last round.
App: gate panel	During drift: the three traffic checks, trust score vs threshold, reputation, and the traffic-only vs full gate decisions.
App: accuracy chart	Running prequential accuracy per arm; stripes = phases; triangles = accepted updates; hover for values.
App: gate chart	Traffic score (green) and reputation (red) with dashed cut-offs; an update is accepted only if both are above their cut-offs.
App: evidence chart	Attribution ('same direction?'), topology ('mixed traffic?'), temporal ('smooth change?') separately.
Live console	One line per round; yellow VETO, green AUTH, blue DRIFT, purple ROLLBACK, red errors; quarantine verdict with its LLR.

12. Testing

108 tests, about 30 seconds (python -m pytest). Tests that need the downloaded datasets skip when they are absent. A mutation check (disabling the veto) makes the headline tests fail, as it should. Adding the quarantine arm is tested to leave the other five arms bit-identical.

File	Covers
test_trust_gate.py	veto, thresholds (inclusive / strict), weight normalisation
test_quarantine.py	SPRT bounds, clamping, stickiness, Monte-Carlo error rates; rollback and replay exactness; commit timing; arm isolation; robustness and lag behaviour
test_pipeline.py	regression against committed synthetic logs, headline result, callbacks, seeds, feature drift, EWC anchoring
test_real_data.py	N-BaIoT shapes and exact reproduction of committed results; IoT-23 parser on hand-written logs; IoT-23 headline and quarantine results
test_reputation.py	noise model: identity, no mutation, bounds, flips, lag
test_fake_nbaiot.py	fake data format, determinism, loader compatibility
test_backend.py	health, SSE streaming, persistence, validation, auth, concurrency limit, quarantine options, log formatting

13. How to run

```
py -3.12 -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
.\.venv\Scripts\python.exe -m pytest

.\.venv\Scripts\python.exe experiments\run_scenarios.py --source real
      [--feature-drift] [--rep-flip 0.3 --rep-sigma 0.05] [--rep-lag 300] [--seed N]
.\.venv\Scripts\python.exe experiments\make_plots.py --tag real
.\.venv\Scripts\python.exe experiments\sweeps.py quarantine --source real

.\.venv\Scripts\python.exe -m uvicorn backend.server:app --port 8765   # console :8765
.\.venv\Scripts\streamlit.exe run app.py                               # app     :8501
.\.venv\Scripts\python.exe docs\build_report.py                        # this PDF
```

Datasets go in data/nbaiot/ and data/iot23/ (gitignored); download instructions are in README.md and PROJECT_GUIDE.txt.

14. Limitations and future work

- The reputation feed is simulated from ground truth; integrating a real threat-intelligence or C2-cost source is Project-II work.
- The SPRT assumes independent per-record feed errors of at most 30%; above that it fails. Correlated errors (a whole blocklist going stale) are not modelled.
- The quarantine hold was raised from 5 to 20 rounds after seeing the lag sweep; other thresholds are untuned defaults.
- Scenario-2 pacing is constructed by interpolation; every endpoint is a real record, but the schedule is imposed.
- On N-BaIoT, seeds only reshuffle calibration rows, so its confidence intervals reflect calibration-order sensitivity, not dataset variability.
- One device pair per dataset; more devices and CICIoT2023 would strengthen the generality claim.
- Future work: learn the feed's reliability online, model correlated feed errors, adaptive hold length from measured feed latency, and adaptive attackers who target the provisional threshold.

References

- Meidan et al., N-BaIoT: network-based detection of IoT botnet attacks using deep autoencoders, IEEE Pervasive Computing, 2018 (UCI dataset 442).
- Garcia, Parmisano, Erquiaga, IoT-23: a labeled dataset with malicious and benign IoT network traffic, Stratosphere Laboratory, 2020.
- Bifet and Gavalda, Learning from time-changing data with adaptive windowing (ADWIN), SDM 2007.
- Wald, Sequential tests of statistical hypotheses, Ann. Math. Stat., 1945.
- Kirkpatrick et al., Overcoming catastrophic forgetting in neural networks (EWC), PNAS 2017.
- Montiel et al., River: machine learning for streaming data in Python, JMLR 2021.
- Rubinstein et al., ANTIDOTE: understanding and defending against poisoning of anomaly detectors, IMC 2009.
- Kravchik et al., Can't boil this frog, arXiv 2002.02741, 2020.
- Korycki and Krawczyk, Adversarial concept drift detection under poisoning attacks, arXiv 2009.09497, 2020.
- Jordaney et al., Transcend, USENIX Security 2017; Yang et al., CADE, USENIX Security 2021; Han et al., OWAD, NDSS 2023.
- Kumar, Fok, Thing, ExpIDS: a drift-adaptable NIDS with improved explainability, arXiv 2509.20767, 2025.
- Nelson et al., Exploiting machine learning to subvert your spam filter (RONI), LEET 2008; Cao et al., FLTrust, NDSS 2021.
- Crab: certified recovery from poisoning in federated learning, arXiv 2401.08216, 2024; AGAT-FL, PeerJ Computer Science, 2025.
- Zhu et al., A distributed sequential algorithm for collaborative intrusion detection networks, arXiv 1002.3190, 2010.